PHYS4055 – Atomic Physics and Quantum Optics – Semester 2, 2026
Assignment 4
Deadline: 4pm Friday 23 October 2026

This assessment contributes 25%/4 towards your overall grade. It consists of *three* questions, on topics covered in the second half of the atomic physics part of the course. Each of the four questions carries equal weight. In answering these questions *show all working.* While you may consult the reference texts and discuss the questions broadly with your peers, you need to carry out and submit *your own work.*

1. Hartree-Fock.

   An important matrix element in atomic structure is that of the two-body Coulomb operator,

   $$\frac{1}{r_{ij}} = \frac{1}{|\mathbf{r}_i - \mathbf{r}_j|} = \sum_{l=0}^{\infty} \frac{r_<^l}{r_>^{l+1}} P_l(\cos\theta) = \sum_{l=0}^{\infty} \sum_{m=-l}^{l} (-1)^m \frac{r_<^l}{r_>^{l+1}} C^l_{-m}(\mathbf{n}_i) C^l_m(\mathbf{n}_j) \ ,$$

   where $r_<$ and $r_>$ is the smaller and greater of the distances r_i and r_j, and P_l is the Legendre polynomial.

   The derivation of the Hartree-Fock equations – that is, the derivation of the direct and exchange Hartree-Fock potentials – involves taking the matrix elements of this equation to evaluate the total energy of the atom. For this, it is necessary to evaluate the two-body matrix element g_{abab} and g_{abba} (using Johnson's notation) which leads to Eqs. (3.64) and (3.66) of Johnson's lecture notes.

   Starting from orbitals given as products of radial, orbital, and spin parts,

   $$\psi_a(\mathbf{r}_i) = (P_{n_a l_a}(r_i)/r_i) Y_{l_a m_a}(\mathbf{n}_i) \chi_{\mu_a}(i) \ , \tag{1}$$

   calculate the Coulomb matrix element and derive the final expressions Eqs. (3.64) and (3.66), showing all working. *If* you are versed in the angular diagram technique, you may use this.

2. Zeeman slower.

   In your answers below, it may be helpful to include the use, or derivation, of equations.

   (a) Explain how the maximum scattering rate influences the design of a Zeeman slower.

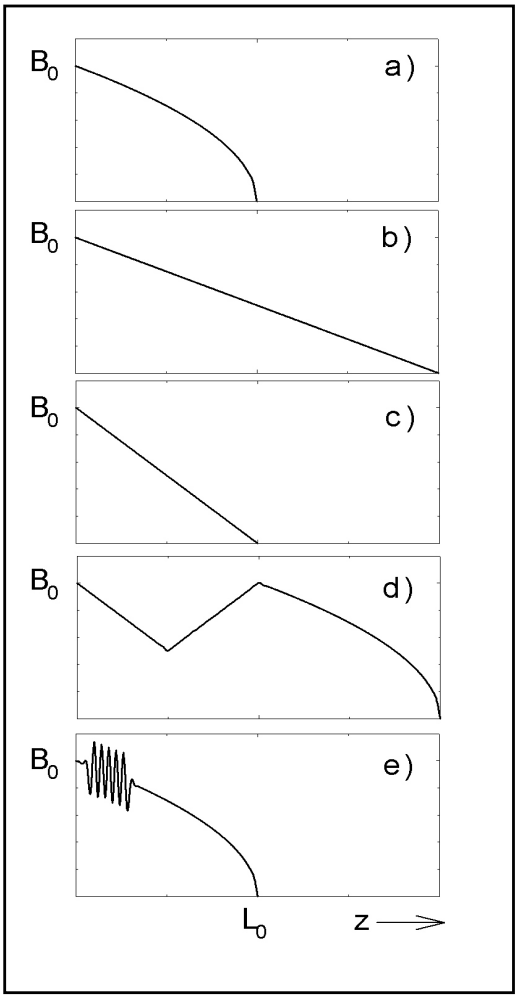
   (b) Explain the shape of the magnetic field required for optimal deceleration efficiency.

   (c) What determines the maximum velocity of the atoms that can still be captured and decelerated by a Zeeman slower?

   (d) The figures below show magnetic fields inside a Zeeman slower that are used to decelerate an atomic beam moving in the z-direction. Panel a) shows the square-root-shaped field which is optimal for decelerating the atoms from their initial velocity down to 0 (at $z = \mathrm{L}_0$). Alternative field shapes are shown in panels b)-e). Discuss whether for fields b)-e) the atoms get decelerated to zero velocity, too. Explain your answers.

3. Lifetimes of atomic states.

   With the aid of the Mathematica Notebook 2 – solving the Schrödinger equation with one of the example parametric potentials – calculate the lifetimes of the states $3p$, $4s$, and $3d$ of the (neutral) Na atom, and compare your results to measured data. Illustrate your working with a sketch of the spectra of the ground state and lowest excited levels, denoting the allowed (E1) transitions. Explain the limitations of your approach.


B0
a)
B0
b)
B0
c)
B0
d)
B0
e)
L0
z